Fantasy Football Performance Analysis

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Introduction/What is Fantasy Football?

- Users act as team managers and draft lineups of real NFL players
- Players earn fantasy points from real-game stats:
  - Passing yards & touchdowns
  - Rushing/receiving yards, touchdowns, receptions (PPR)
  - Defensive & kicking statistics
- Fantasy performance is shaped by projections, week-to-week performance variability, injuries/bye weeks , usage changes and aging effects
- Our project analyzes positional consistency, projection accuracy and key factors affecting performance

Data & Cleaning

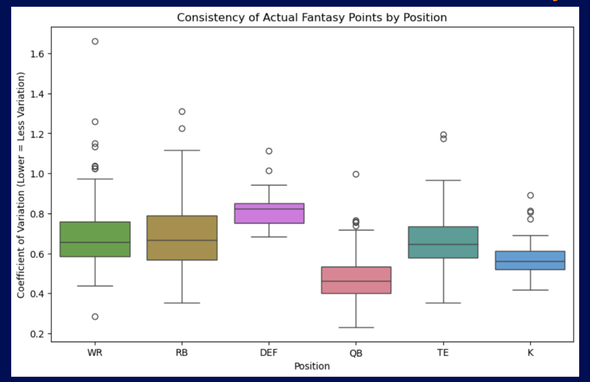
- Data source: Sleeper API
- Includes player info, weekly stats, actual & projected PPR points from 2018-2024
- Filtered out players with < 5 projected points
- Final dataset: ~30,000 rows

Research Questions

1. Which positions contributed the most consistent fantasy points?
2. Which players outperform projections most often?
3. Are there seasonal trends in fantasy scoring?
4. How are players affected off an absence?
5. How well does previous season points project next season's fantasy points?
6. How do projects adjust for performance from the previous week?
7. How does age affect performance & projections?

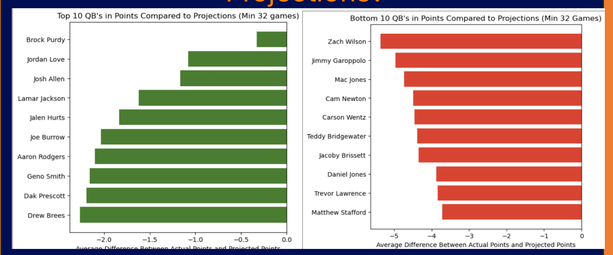
Results

Which Positions Perform Most Consistently?

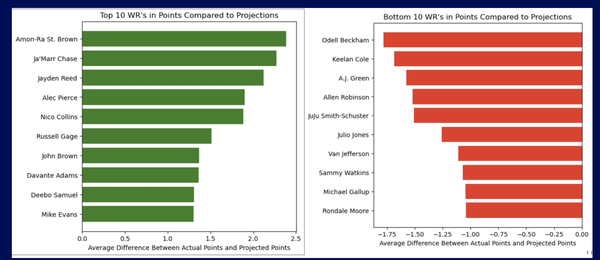


- QB's are the most consistent performers
- Defense performance varies the most, but variation is pretty consistent across all defenses.
- WR, RB, TE performance varies the most.

Which Players Over/Underperform Projections?

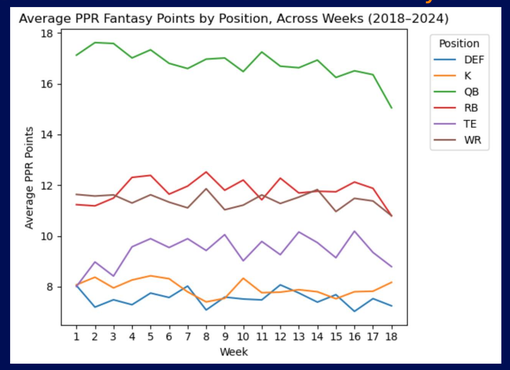


- Every QB underperforms projections on average.
- Elite QB's underperform the least.
- Notorious busts and players near the end of their careers underperform the most.



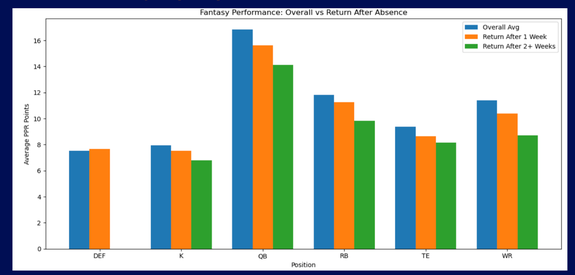
- RB, WR, and TE follow this pattern:
  - Young stars overperform the most.
  - Older, past their prime players underperform the most
- There is no pattern for K or DEF, other than projections understating the greatness of Brandon Aubrey.

Are there Seasonal Trends in Fantasy Scoring?



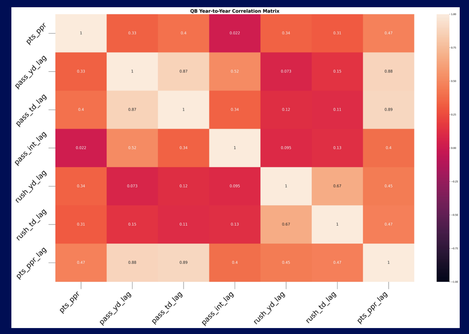
- Performance overall stays consistent throughout the season.
- Dips at the end due to wear and tear or teams resting starters.

How does Injury/Bye Weeks Affect Performance?



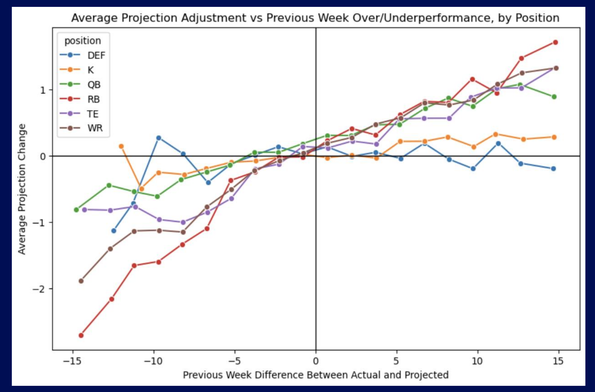
- Offensive performance is worse off an absence (Bye week) and gets worse with 2+ week absences (Injuries).
- The bye weeks seems to be beneficial for defensive units.

How Well does Previous Season Peformance Project Next Season Performance?



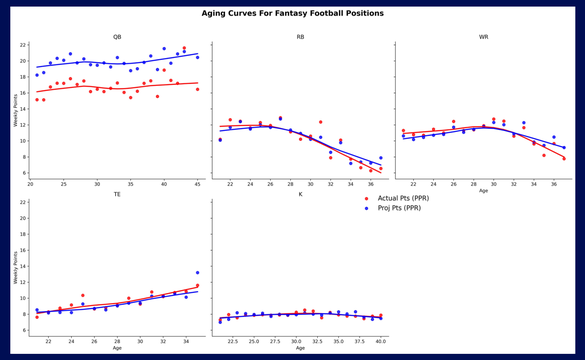
- Last season fantasy points has most significant impact on future points
- QB Passing TDs are the most consistent from year-to-year
- Yards gained are the most important for RBs/WRs/TEs
- Ks/DEFs don't show much consistency or trends from year-to-year

How do Projections Adjust for Over/Underperformance in the Previous Week?



- QB, RB, WR, and TE projections all increase with a good performance in the previous week.
- K and DEF projections are more matchup-based than performance-based.

How does Age Affect Performance & Projections?



- Age doesn't seem to affect QBs, TEs, or Ks very significantly
- RBs tend to peak at 26-28 before starting to decline
- WRs Peak at around 30 before declining

Main Takeaways

- Projection systems seem to do a poor job at projecting QB performance, but QBs also provide the most consistent performance from week-to-week.
- Fantasy performance remains consistent as the season progresses across all positions
- Bye weeks & injuries seem to slightly hurt performance in every position besides defense
- Across all positions, last season fantasy points does the best job of predicting future performance
- Projections are more performance-based for WRs & RBs while DEF are more matchup based
- RBs & WRs see performance decline by their 30s while QBs & Ks aren't affected by age as much